Peer Effects Among Inventors: Unpacking Their Origins in Cancer-Related Innovation

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1 Introduction

Theories of knowledge spillovers have been central to economic thought since Marshall [1890] first emphasized the productivity advantages firms derive from geographic concentration in industrial districts. Marshall's insights laid the groundwork for understanding how proximity facilitates the exchange of ideas and skills, enhancing collective efficiency. Building on this foundation, contemporary research has extended the concept of knowledge spillovers into diverse domains such as industrial innovation, geographic agglomeration, economic growth, and international trade. In particular, a growing body of literature highlights that the transmission of knowledge is not merely a byproduct of proximity, but is increasingly shaped by collaboration—both among private firms and between public and private institutions—which plays a crucial role in driving innovation and technological progress [Cowan and Jonard, 2004, Sorenson et al., 2006].

Knowledge production today is increasingly the result of interactions within intra- and inter-organizational networks, where inventors from diverse fields of expertise collaborate, exchange tacit knowledge, and collectively drive technological research [Grigoriou and Rothaermel, 2014, Jaravel et al., 2018, Moreira et al., 2018, Nerkar and Paruchuri, 2005, Paruchuri, 2010]. These networks not only enable collaboration and mutual learning but also foster enduring professional relationships that enhance individual problem-solving capacity and the innovative potential of their organizations. The knowledge exchanged within these networks, often mediated by social capital, helps explain persistent performance differences among inventors and firms. These disparities are frequently rooted in the structure and strength of collaborative ties, which lie at the heart of the knowledge spillover process [Aral and Van Alstyne, 2011, Hansen, 2002, Carnabuci and Operti, 2013, Fleming et al., 2007]. Despite growing attention to collaborative dynamics, the specific mechanisms by which knowledge is created and transmitted remain only partially understood. Much of the innovation literature continues to focus on firm-level R&D investments and inter-firm knowledge transfers as primary drivers of technological progress [Bernstein and Nadiri, 1989, Bloom et al., 2013, Hsieh et al., 2024, König et al., 2019, König and Rogers, 2023]. While human interaction and spatial proximity are often cited as key drivers of knowledge exchange, empirical studies frequently rely on aggregate production data. This reliance limits researchers' ability to directly test the micro-level processes involved, leaving important gaps in our understanding of how knowledge generation actually occurs in practice.

This paper is the first to quantify heterogeneity in peer effects among inventors by examining how network position and individual characteristics jointly influence inventive productivity. The microfoundations build on the model proposed by Liu et al. [2014], in which peer effects arise through two distinct behavioral mechanisms: strategic complementarity and conformity. The analysis rests on two key premises. First, an inventor's productivity can be decomposed into an idiosyncratic component and a peer-driven component—reflecting both individual attributes and the influence of collaborators. Second, at the group level, peer effects capture the mutual and reciprocal influence among inventors, encompassing both aggregate and bilateral dynamics within collaboration networks. This framework formally links productivity to an inventor's network position, using a variant of the centrality measure introduced by Bonacich [1987].

I then examine the econometric counterpart of this theoretical model using two widely adopted specifications in the spatial econometrics literature: the local-average and local-aggregate models. The key distinction between them lies in the treatment of the adjacency matrix representing collaboration: the local-average model employs a row-normalized matrix, whereas the local-aggregate model does not. The underlying game-theoretic framework provides microfoundations for both specifications, offering a behavioral interpretation of the structural assumptions embedded in each approach [Liu et al., 2014].

Within this framework, the main parameters of interest capture not only the strength but also the behavioral origins of dyadic influence in the network. These parameters are crucial in the computation of Katz-Bonacich centrality, where they function as decay factors that weight paths by length and differentiate between the influence of strategic complementarity and conformity. Consequently, they help identify both the extent and origin of peer effects among inventors. However, estimating this model introduces several empirical challenges.

The main challenge lies in simultaneously accounting for the effects of collaboration and the matching process that governs which inventors form ties. At the dyadic level, greater inventive output is often associated with more intense collaborative effort. Since each inventor productivity is not constant, it is essential to model the idiosyncratic productivity in order to get a credible measure of the network effects. Moreover, co-inventorship is not randomly assigned—it typically arises from shared research interests and strategic considerations. An inventor's decision to collaborate, and hence their position within the broader network, is often shaped by expected returns to collaboration. As a result, the network structure is endogenously determined, influenced by strategic choices and shaped by unobserved factors that affect both productivity and the propensity to collaborate [Mairesse and Turner, 2006, Fafchamps et al., 2010, Freeman et al., 2014]. To address this challenge, I implement an instrumental variable approach developed by Lee et al. [2021], which uses logistic regression to predict the probability of link formation between all inventor pairs based on exogenous dyadic characteristics—such as institutional affiliation, demographic similarities, and other relevant factors. The predicted network is then used as an instrument. This method remains robust in settings where network connections are endogenously formed and reduces computational complexity by avoiding the need to explicitly model the entire network formation process.

I test the proposed peer effects model using the *Cancer Moonshot patent data* compiled by the USPTO [Frumkin and Myers, August 2016], which focuses on cancer-related inventions. This dataset is enriched by linking it to the disambiguated inventor database from Li et al. [2014], along with additional metadata provided by the Torvik research group. Cancer research offers a particularly relevant context due to its strong reliance on collaboration and the transfer of tacit knowledge—best facilitated through direct interaction [Polanyi, 1966, Nonaka and Takeuchi, 2007]. Moreover, cancer-related innovation has substantial economic implications; for example, a 1% annual reduction in mortality from major cancers (lung, colorectal, breast, leukemia, pancreatic, and brain) is estimated to reduce productivity costs by $814 million annually [Bradley et al., 2008].

The first empirical finding of this study supports the theoretical model: Bonacich centrality has a significant positive effect on scientific productivity. Peer effects among inventors operate through both strategic complementarity and conformity to immediate collaborators. In equilibrium, inventors with the highest centrality are also those with the highest output. This result is robust to various controls, including seniority, past productivity, and gender. These findings highlight the importance and magnitude of network externalities in shaping inventive performance.

A second key finding is that, among standard network measures, Bonacich centrality stands out due to its microfounded, dyadic, and global characteristics. Comparing its explanatory power with that of other commonly used centrality metrics provides strong empirical support for the proposed microfoundation. The Nash equilibrium characterization embedded in Katz-Bonacich centrality offers valuable analytical benefits, with important implications for comparative statics and the design of optimal network policies [Ushchev and Zenou, 2020].

This study contributes to innovation and strategic management literature in several ways. First, it fills a gap on how co-inventorship networks affect individual productivity and invention quality, an area underexplored despite extensive research on peer effects in academic knowledge production [Azoulay et al., 2010, 2019, Borjas and Doran, 2012, Mohnen, 2022, Waldinger, 2010, 2012, Oettl, 2012]. Second, while R&D spillovers across firms—especially within sectors or geographic proximity—are well documented [Jaffe, 1989, Jaffe et al., 1993, Cohen and Levinthal, 1989], collaborative ties often amplify these effects [Branstetter and Sakakibara, 2002, Gomes-Casseres et al., 2006]. Firm network position significantly influences innovation outcomes [Powell et al., 1996, Ahuja, 2000, Schilling and Phelps, 2007, König et al., 2019], yet the role of individual inventors remains understudied. This work complements Konig et al. [2023], Zacchia [2020] by providing a behavioral foundation for inventor-level peer effects and highlighting social conformity and strategic complementarity as key mechanisms. Third, related research emphasizes the role of teams, cognitive diversity, and co-mobility in driving innovation [Agrawal et al., 2008, Alexander and Van Knippenberg, 2014, Bhaskarabhatla et al., 2021, De Dreu, 2006, Jones, 2009, Groysberg and Lee, 2009, Campbell et al., 2014, Palomeras and Melero, 2010], while formal models elucidate knowledge spillovers across inventors [Stein, 2008, Lucas Jr and Moll, 2014].

The paper proceeds as follows. Section 2 introduces the game-theoretical framework, modeling peer-effects within inventor networks. Section 3 presents the econometric framework and empirical strategy. Section 4 describes the data construction and sample selection. Section 5 reports the empirical findings. Section 6 concludes.

2 Micro-foundations at the inventor level

The primary question examined in this paper is whether the effects of peer interactions on inventive output—specifically, patent production—are driven by *strategic complementarity*, *social conformity*, or a combination of both. To explore this, I draw upon the peer effects model developed by Liu et al. [2014], which effectively captures the interaction of these two mechanisms and allows for a broader understanding of their impact on innovation. This model integrates both *local-average* and *local-aggregate* peer effects models, which are useful in studying how the behavior of one inventor might influence the patent output of their peers.

The model relies on two key assumptions: first, that inventors benefit from the increased patent output of their collaborators, and second, that they experience a cost when their output deviates from the prevailing norms within their social or professional network. In equilibrium, an inventor's output is determined by their position in the network, where their incentives are shaped both by the choices of their co-inventors (through strategic complementarity and conformity) and by their own individual productivity.

2.1 The Network and Notations

Consider a set of inventors $N = \{1, \ldots, n\}$ grouped into distinct communities, with each community denoted as $N_c = \{1, \ldots, n_c\}$. The collaboration between inventors within each community is captured by an adjacency matrix $G_c = [g_{ij,c}]$, where $g_{ij,c} = 1$ indicates that inventors i and j have collaborated on one or more patents, and $g_{ij,c} = 0$ otherwise. Additionally, $g_{ii,c} = 0$, meaning an inventor does not collaborate with themselves. Each inventor i has a set of co-inventors $N_{i,c} = \{j \mid g_{ij,c} = 1\}$, and the number of co-inventors is represented by the degree $|N_{i,c}|$. Let $y_{i,c}$ denote the patent output of inventor i in community c. The aggregate patent output across the community is represented by $\mathbf{Y}_c = (y_{1,c}, \ldots, y_{n_c,c})$.

2.2 Modeling the Benefits of Collaboration

Theories of knowledge spillovers have been central to economic analysis since Marshall [1890], who highlighted their role in explaining the productivity advantages firms experience by clustering together in manufacturing districts of 19th-century England. A substantial body of literature has since argued convincingly that knowledge is not simply "in the air" and accessible to all agents. Instead, it follows specific channels between individuals who are socially and personally connected [Breschi and Lissoni, 2005, 2009, Knoben, 2009]. At the core of our theoretical framework on technological peer effects is the idea that technological production is inherently cooperative. An inventor's success is shaped by their ability to collaborate with others, and this cooperation takes various forms. One key element of this collaboration involves the exchange of tacit knowledge. Inventor collaboration, or coinventing, requires extensive communication, not only in the form of verbal and nonverbal interactions [Allen and Cohen, 1969, Singh, 2005] but also through the mutual sharing of expertise.

Consequently, inventor collaboration ties can be seen as nondirectional communication channels [Allen, 1977, Nerkar and Paruchuri, 2005]. It follows that well-connected inventors are better positioned to access useful knowledge and leverage the skills of their co-inventors, making them more productive. In the context of innovation, peer effects are essential for understanding how an inventor's productivity is influenced by their network of collaborators. *Strategic complementarity* posits that an inventor's patent output is positively influenced by the productivity of their co-inventors. Collaboration offers inventors opportunities to share ideas, refine methodologies, and provide feedback—all of which can enhance individual performance. Collaborating with high-output inventors, in particular, exposes them to valuable knowledge and best practices, which further boosts their productivity.

This dynamic implies that inventors who collaborate with more productive peers are likely to see an increase in their own patent output. Thus, an inventor's total output is determined not only by their own productivity but also by the combined output of their co-inventors. To capture this, the model expresses an inventor's patent production as a function of both their own output and the output of their co-inventors:

$$\underbrace{\left(\pi^{\star}_{i,c} + \lambda_1 \sum_{j=1}^{n_c} g_{ij,c} y_{j,c}\right) y_{i,c}}_{\text{benefits from collaboration}},$$

where $\pi^{\star}_{i,c}$ represents an inventor's idiosyncratic productivity, and $\sum_{j \in N_{i,c}} y_{j,c}$ is the total output of their co-inventors. The parameter $\lambda_1 \geq 0$ captures the strength of strategic complementarity, meaning that the higher the productivity of co-inventors, the greater the incentive for an inventor to increase their own output.

2.3 Modeling the Cost of Deviation from the Norm

Collaboration offers significant advantages, but it also comes with certain costs, particularly due to the social dynamics within inventor networks. One such cost arises from social conformity, where inventors may face penalties if their output significantly deviates from the average output of their peers. In other words, inventors are not only concerned with their own productivity but also with maintaining alignment with the collective norms of their peers. This desire for conformity may lead inventors to adjust their output to match the behavior of others, thereby minimizing the costs of deviating from the group's norm. Specifically, Kandel and Lazear [1992] suggest that peer pressure can be an important driver of outcome in a team work environment. They explore how peer pressure, along with factors such as social norms and mutual monitoring, creates incentives for individuals to contribute more. Another potential mechanism fostering interdependence is relative compensation. When an individual's chances of promotion or job security depend on their performance relative to that of their peers, they may increase their effort in response to higher productivity from others. The cost of deviating from the group norm is modeled as a function of the difference between an inventor's output and the average output of their co-inventors:

$$\underbrace{\frac{1}{2}\left[y_{i,c}^2 + \lambda_2\left(y_{i,c} - \sum_{j=1}^{n_c} g_{ij,c}^{\star} y_{j,c}\right)^2\right]}_{\text{cost of deviation}},$$

where $\lambda_2 \geq 0$ reflects the intensity of the conformity effect, meaning the larger the deviation from the reference group's average output, the higher the cost.

2.4 The Peer Effects Game

In this model, inventors in a community c simultaneously decide how much to produce in terms of patents, aiming to maximize their individual utility. The utility function for inventor i is given by:

$$\begin{aligned} u_{i,c}(y_{i,c}) \equiv u_{i,c}\left(y_{i,c}, \mathbf{Y}_c, \mathbf{G}_c\right) = & \underbrace{\left(\pi_{i,c}^{\star} + \lambda_1 \sum_{j=1}^{n} g_{ij,c} y_{j,c}\right) y_{i,c}}_{\text{benefits}} \\ & - \underbrace{\frac{1}{2}\left[y_{i,c}^2 + \lambda_2 (y_{i,c} - \sum_{j=1}^{n} g_{ij,c}^{\star} y_{j,c})^2\right]}_{\text{cost}}. \end{aligned} \tag{1}$$

This setup captures the decision-making process of inventors within a network, where their utility is influenced by both the benefits of collaboration and the costs of deviating from the social norms of their community.

The best-reply function for inventor i, derived from maximizing the utility function, is given by:

$$y_{i,c} = \phi_1 \sum_{j=1}^{n_c} g_{ij,c} y_{j,c} + \phi_2 \sum_{j=1}^{n_c} g_{ij,c}^{\star} y_{j,c} + \pi_{i,c},$$

where $\phi_1 = \frac{\lambda_1}{1+\lambda_2}$ and $\phi_2 = \frac{\lambda_2}{1+\lambda_2}$ are the coefficients that reflect the relative importance of strategic complementarity and social conformity, respectively.

I denote by $g_c^{\max}$ the highest degree in network c, i.e. $g_c^{\max} = \max_i g_{i,c}$. Let $\mathbf{\Pi}_c = (\pi_{1,c}), \cdots, \pi_{nc,c}$. The Nash equilibrium of the general network model is characterized by the following proposition:

Proposition 1 [Liu et al., 2014]

If $\phi_1 \geq 0$, $\phi_2 \geq 0$, and $g_c^{\max}\lambda_1\lambda_2 < 1$, then the network game with payoffs (1) has a unique interior Nash equilibrium in pure strategies given by:

$$\mathbf{Y}_c = \left(\mathbf{I}_{nc} - \phi_1 \mathbf{G}_c - \phi_2 \mathbf{G}_c^{\star}\right)^{-1} \mathbf{\Pi}_c$$

In equilibrium, each inventor's productivity is proportional to a variation Katz-Bonacich centrality [Bonacich, 1987], reflecting the recursive influence structure embedded in the network. We are dealing with a game with strategic complementarities, so the higher the production of coinventors, the higher the marginal utility of raising my own production. Consequently, there is a problem of equilibrium existence since there is no bound on the production of each inventor. Since ϕ_1 and ϕ_2 express the intensity of these complementarities, the condition $g_c^{\max}\phi_1\phi_2 < 1$ guarantees the existence of equilibrium by bounding the relative strength of strategic complementarities of production with respect to the amplifying power of the network. The linear best-reply functions combined with strategic complementarities rule out the multiple equilibria case.

2.5 Equilibrium and Policy Implications

In equilibrium, inventors' outputs are determined by their interactions within the network. The model predicts that in a conformity-dominant environment (i.e., $\lambda_1 = 0$), inventors will adjust their patent output to match the average production level of their co-inventors. In contrast, in a strategic complementarity-dominant environment (i.e., $\lambda_2 = 0$), the total output of an inventor's collaborators directly influences their own output, meaning that inventors in more productive networks will tend to produce more patents themselves.

The policy implications differ significantly based on which mechanism predominates. If social conformity drives patent output, policy interventions aimed at changing the social norms of a group (e.g., creating new standards or incentives) would be necessary. On the other hand, if strategic complementarity is the key driver, then targeting high-productivity inventors with individual-based policies (such as bonus based on the output) could amplify innovation throughout the network.

3 Empirical strategy

This section extends the work of Liu et al. [2014], proposing a version that accommodates endogenous interactions within the network and incorporates heteroscedastic errors.

3.1 Econometric Model of the Network

The econometric model is grounded in equilibrium behavior derived from a network game framework, providing a clear microfoundation. Specifically, we account for ex ante heterogeneity of each inventor i in community c using the following equation:

$$\pi_{i,c} = \eta_c + x_{i,c}^{\intercal}\boldsymbol{\beta_1} + u_{i,c},$$

where $x_{i,c}$ represents a vector of exogenous variables. The error term $u_{i,c}$ is assumed to be heteroscedastic across individuals, with the covariance matrix $\Sigma = \text{diag}(\sigma_i^2)$, indicating that the errors are independent across individuals. The vectors $\boldsymbol{\beta_1}$ contains the parameters to be estimated.

The full econometric network model can be written as:

$$y_{i,c} = \phi_1 \sum_{j=1}^{n_c} g_{ij,c} y_{j,c} + \phi_2 \sum_{j=1}^{n_c} g_{ij,c}^{\star} y_{j,c} + \eta_c + x_{i,c}^{\intercal}\boldsymbol{\beta_1} + u_{i,c},$$

where $i = 1, \ldots, n_c$ and $c = 1, \ldots, \bar{c}$. The vectors $\mathbf{Y}_c = (y_{1,c}, \ldots, y_{n_c,c})^{\intercal}$, $\mathbf{X}_c = (x_{1,c}, \ldots, x_{n_c,c})^{\intercal}$, and $\mathbf{u}_c = (u_{1,c}, \ldots, u_{n_c,c})^{\intercal}$ are the vectors of outcomes, exogenous variables, and error terms, respectively. In matrix form, the model becomes:

$$\mathbf{Y}_c = \phi_1 \mathbf{G}_c \mathbf{Y}_c + \phi_2 \mathbf{G}_c^{\star} \mathbf{Y}_c + \eta_c \boldsymbol{\iota}_{n_c} + \mathbf{X}_c \boldsymbol{\beta_1} + \mathbf{u}_c.$$

To model the system across multiple communities, we define $\mathbf{Y} = (\mathbf{Y}_1^{\intercal}, \ldots, \mathbf{Y}_{\bar{c}}^{\intercal})^{\intercal}$, and similarly for $\mathbf{X}$, $\mathbf{u}$, and $\boldsymbol{\eta}$. This results in the system:

$$\mathbf{Y} = \phi_1 \mathbf{G}\mathbf{Y} + \phi_2 \mathbf{G}^{\star}\mathbf{Y} + \mathbf{X}\boldsymbol{\beta_1} + \mathbf{L}\boldsymbol{\eta} + \mathbf{u}.$$

Here, $\mathbf{L}$ is a block-diagonal matrix representing community-specific effects. To eliminate the community-specific heterogeneity and address the incidental parameter problem, I use the within-transformation, where $\mathbf{J}_c = \mathbf{I}_{n_c} - \frac{1}{n_c}\boldsymbol{\iota}_{n_c}\boldsymbol{\iota}_{n_c}^{\intercal}$ projects the variables onto their deviations from the community means. The transformed model becomes:

$$\mathbf{J}_c\mathbf{Y} = \phi_1 \mathbf{J}_c\mathbf{G}_c\mathbf{Y} + \phi_2 \mathbf{J}_c\mathbf{G}_c^{\star}\mathbf{Y} + \mathbf{J}_c\mathbf{X}_c\boldsymbol{\beta_1} + \mathbf{J}_c\mathbf{L}\boldsymbol{\eta} + \mathbf{J}_c\mathbf{u}. \tag{2}$$

The model outlined in Equation (2) consists of two key components: endogenous effects and idiosyncratic effect. The endogenous effects, reflected by ϕ_1 and ϕ_2, capture how an individual's outcome is influenced by the outcomes of her coauthors in the network, while the idiosyncratic effect captures the influence of observables characteristics on the productivity. Additionally, the model distinguishes between, aggregate endogenous effects (ϕ_1) and the average endogenous effects (ϕ_2), each providing different insights into the equilibrium outcomes depending on the structure of the network.

3.2 Estimation Approach

The goal is to consistently estimate the parameters $\theta = (\phi_1, \phi_2, \beta_0, \beta_1^{\intercal})^{\intercal}$. The challenge lies in addressing potential endogeneity and biases in the network data.

3.2.1 Identification Challenges

The estimation of this model is not without any challenges, as emphasised by the seminal work of Manski [1993]. The technical difficulties around peer effect estimation have been widely studied and have customarily been divided into reflection bias, selection bias, and unobserved correlated effects [Brock and Durlauf, 2001, Moffitt et al., 2001]. There are several challenges in identifying the parameters of Equation (2). First, the so-called feedback-loops between inventors imply that an inventor's behavior may influence the behavior of her collaborator and vice versa, leading to simultaneity bias. Second, the non-random sorting of inventors into communities—based on observable or unobservable factors—may induce selection bias. Third, endogenous collaboration, which form based on anticipated returns from collaboration rather than being random, may lead to confounding in the estimation of peer effects.

Correlated or common-shock problems arise in network models due to the fact that there may be common environmental factors that cause the inventors in the same community to behave in a similar manner. This may be confounded with the co-inventors-effect ϕ_1 and ϕ_2 I want to identify. One way of dealing with this issue is by introducing community fixed effects.

In fact, the structure of interaction helps shape inventors' productivities, as highlighted by the model outlined earlier. The very determination of the co-invention links upon which these outcomes are resolved is plausibly informed by these outcomes or by expectations of them in many cases. It cannot be assumed that the peer structure, i.e. the links between the different individuals involved, is not determined independently of expectations about outcomes. If there were factors that affect co-inventorship and inventor productivity, but these were not properly specified in the model, elements of the adjacency matrix would correlate with the disturbance term of the interaction model. For instance, highly creative inventors may be sorted with other highly creative inventors, as creativity levels may affect not only their patenting outcomes, but also the collaboration, an unobservable factor that affects them simultaneously.

These challenges are similar to what Manski [1993] terms *correlated effects*, where the observed similarities in outcomes among coinventors may stem from pre-existing similarities rather than true peer effects. The endogenous formation of coinventorship ties introduces a bias that could distort the estimation. Addressing these concerns requires careful attention to both the model specification and the chosen estimation strategy. In our cases, I argue that sharing the same ethniciity is a valid exclusion restriction . I assume that sharing a common cultural background is exogenous to the current production process, by being at the same times a predictor of the existence of ties between inventors. To assess the relevance of our instrument, I show that sharing a common cultural background. Showing that the exclusion restriction holds is, of course, far more difficult. The key assumption is that sharing a same cultural background only influences the intensity of the collaboration and not the output over the period.

Although plausible, in this case we believe that this dimension is purged by adding community-level fixed effects. One general limitation, of course, is that this may not help with unobservables that are not common. Potential problems of omitted variables are thus reduced but not eliminated.

3.2.2 Endogeneity and instrumental variables

When the adjacency matrix $\mathbf{G}$ is assumed to be exogenous, linear social interaction models of the form of Equation (2) can be estimated using two-stage least squares (2SLS) based on the moment condition:

$$\mathbf{Z}^{\intercal} u(\theta) = 0,$$

where $\mathbf{Z}$ is the instrumental variables (IV) matrix composed of the linearly independent columns from the set $\left[\mathbf{J}[\mathbf{X}, \mathbf{G}^{\star}\mathbf{X}, \mathbf{G}\mathbf{X}, \mathbf{G}^{\star 2}\mathbf{X}]\right]$. We also define $\underline{\mathbf{X}} = \mathbf{J}[\mathbf{G}\mathbf{Y}, \mathbf{G}^{\star}\mathbf{Y}, \mathbf{X}]$ and the equation for the residuals becomes:

$$\mathbf{J}\mathbf{u}(\theta) = \mathbf{J}\mathbf{Y} - \underline{\mathbf{X}}\theta.$$

The parameters in Equation (2) can be consistently identified, provided that a certain level of intransitivity exists in the network such that the usual rank condition for IV estimators holds [Bramoullé et al., 2009, Liu and Lee, 2010, Liu et al., 2014].

However, if there exists an unobserved individual factor affecting both the output and the collaboration between inventors, then $\mathbf{G}$ becomes endogenous, and the IV matrix $\mathbf{Z}$ is no longer valid. In this case, the 2SLS method can be adjusted by replacing the observed adjacency matrix $\mathbf{G}$ in the IV matrix with a predicted adjacency matrix $\hat{\mathbf{G}}$, which is based on exogenous covariates [Kelejian and Piras, 2014].

For each pair of individuals i and j in the network, the utility that individual i derives from forming a link with individual j can be expressed as:

$$U_{ij} = \delta_0 + W_{ij}\delta_1 + \epsilon_{ij},$$

where U_{ij} is the utility for inventor i from collaboration with inventor j, W_{ij} is the deterministic component of the utility, dependent on observable characteristics of both i and j, and ϵ_{ij} is the stochastic component capturing unobserved factors influencing the collaboration. Assuming ϵ_{ij} follows an i.i.d. Type-I extreme value distribution, I obtain the following logistic regression model for dyadic network formation:

$$\mathbb{P}(g_{ij} = 1) = \frac{\exp(\delta_0 + W_{ij}\delta_1)}{1 + \exp(\delta_0 + W_{ij}\delta_1)}. \tag{3}$$

From this, the predicted link formation probability is:

$$\hat{g}_{ij} = \frac{\exp(\hat{\delta}_0 + W_{ij}\hat{\delta}_1)}{1 + \exp(\hat{\delta}_0 + W_{ij}\hat{\delta}_1)},$$

where $\hat{\delta}_0$ and $\hat{\delta}_1$ are consistent estimators. To ensure the predicted adjacency matrix is uniformly bounded in both row and column sums, I normalize $\hat{g}_{ij}$ by dividing it by $\hat{d} = \max\left(\max_i \sum_{j=1}^{n} \hat{g}_{ij}, \max_j \sum_{i=1}^{n} \hat{g}_{ij}\right)$ [Kelejian and Prucha, 2010]. This results in a normalized predicted adjacency matrix $\hat{\mathbf{G}}$, where:

$$\hat{g}_{ij}/\hat{d} \quad \text{if} \quad i \neq j, \quad \text{and} \quad 0 \quad \text{otherwise.}$$

The IV matrix, based on the predicted adjacency matrix $\hat{\mathbf{G}}$, is denoted by $\hat{\mathbf{Z}}$, and it includes linearly independent columns from:

$$\mathbf{J}[\mathbf{X}, \hat{\mathbf{G}}\mathbf{X}, \hat{\hat{\mathbf{G}}}\mathbf{X}, \hat{\hat{\mathbf{G}}}^2\mathbf{X}, \hat{\mathbf{G}}\mathbf{1}].$$

The corresponding 2SLS estimator is then given by:

$$\hat{\theta}_{2sls} = \left[\underline{\mathbf{X}}^{\intercal}\hat{\mathbf{Z}}\left(\hat{\mathbf{Z}}^{\intercal}\hat{\mathbf{Z}}\right)^{-1}\hat{\mathbf{Z}}^{\intercal}\underline{\mathbf{X}}\right]^{-1}\underline{\mathbf{X}}^{\intercal}\hat{\mathbf{Z}}\left(\hat{\mathbf{Z}}^{\intercal}\hat{\mathbf{Z}}\right)^{-1}\hat{\mathbf{Z}}^{\intercal}\mathbf{J}\mathbf{y}.$$

As pointed out by Lee et al. [2021], the consistency of the proposed 2SLS estimator does not rely on the consistency of the estimator $\hat{\delta} = (\hat{\delta_0}, \hat{\delta_1}^{\intercal})^{\intercal}$.

4 Patent data, sample construction

Five parts composed this section. The first and the second introduce datasets used to build our database, but also their limitations. In the third part, we document the construction of the star inventor networks in our study. Finally, I document the construction of the variables and their descriptive statistics.

4.1 Data Sources

Patents The data used in this study were compiled by the USPTO Cancer Moonshot task force. This database includes patent information related to cancer research from 1976 to 2016, covering all patents granted by the USPTO as well as all patent applications. In addition to the specific patent data, the database is enriched with information from several other sources: grants awarded by the National Institutes of Health (NIH), data from the Research Portfolio Online Reporting Tools (RePORTER), and details from the U.S. Food and Drug Administration's (FDA) Approved Drug Products with Therapeutic Equivalence Evaluations, commonly referred to as the Orange Book.

Inventor Name disambiguation A persistent challenge in collecting inventor-specific patent data is the issue of name disambiguation, as patent databases typically lack unique identifiers for individual inventors. However, the USPTO Dataset available through patentsview.org addresses this problem by providing disambiguated inventor and assignee names. This dataset covers all patents granted between 1976 and 2024 and builds upon the methodology developed by Li et al. [2014], which applies name disambiguation techniques to probabilistically assign unique identifiers to inventors. This approach is becoming a foundational tool for numerous studies focused on identifying individual inventors and analyzing their productivity within the USPTO patent data. In addition to disambiguating inventor names, Li et al. [2014] also link disambiguated inventors to author identities in the PubMed database.

Inventor Attributes To complement these two datasets, I use metadata compiled by the Torvik research group and its various projects to construct inventor-level variables, specifically their gender and ethnicity. The gender of each inventor is determined using a tool called Genni, which assigns a gender to a first name through a probabilistic model that incorporates the ethnicity linked to the inventor's last name. For instance, the first name Andrea will be classified as female if the surname is associated with the French ethnicity, but as male if the surname is linked to the Italian ethnicity.

The Ethnea tool[1] enables the matching of ethnicity to an inventor based on their first and last names. Unlike some other ethnicity-matching algorithms, Ethnea offers a contemporary perspective on ethnicity, focusing more on nationality rather than distant ancestry. This approach also accounts for dual ethnicities, which are increasingly common due to factors such as marriage, migration, and cultural assimilation.

To identify an inventor's employer, I use data on ownership reassignments from the USPTO's patent assignment database. Before September 2012, the USPTO initially considered the inventor to be the legal owner of a patent application. However, because inventors are typically contractually obligated to assign their rights to their employer, these transfers had to be formally recorded through a reassignment process [Marco et al., 2015]. For inventors who also published scientific articles during the study period, I supplement this approach by using the MapAffil tool to match them to their affiliated institutions.

4.2 Data limitations

There are two primary limitations associated with the data. First, patent counts serve as an imperfect proxy for the innovation produced by an inventor. Not all patents are of equal value—some represent significant technological advances, while others are incremental or even trivial. In certain cases, patents are filed for strategic or defensive purposes, without embodying genuine innovation. To address this limitation, I estimate models in which the dependent variable is the number of patents per inventor, weighted by a breakthrough index. In particular, I adopt the approach of Kelly et al. [2021], who employ textual analysis of patent documents to construct a high-dimensional measure of technological innovation. An interesting feature of this measure is that it predicts future citations and correlates with patent market value as estimated by Kogan et al. [2017], while not being limited to patents granted to publicly traded firms.

Second, not all innovations are patented, and the propensity to patent varies across industries and technological domains. For instance, Cohen et al. (2003) report that in industries such as food, textiles, steel, and other metals, firms patent fewer than 15% of their product innovations. However, in the domains relevant to this study—specifically chemical, pharmaceutical, and medical device industries—more than two-thirds of innovations are patented. Within this context, patent data can be considered a reliable proxy for innovative activity and collaboration. Moreover, patent records provide an accurate picture of the collaborative structure underlying innovation, as legal requirements stipulate that all individuals who have made a significant contribution must be named as inventors, and only those meeting this criterion are permitted to be listed.

4.3 Network construction

To prepare the data for estimation, I implement the following sample selection procedure. First, the analysis is restricted to patents filed within three-year periods to ensure consistency in temporal comparisons. Second,

[1] Ethnea" tool from the University of Illinois establishes the ethnicity of inventors based on their first and last names. This tool is available at http://abel.lis.illinois.edu/cgi-bin/ethnea/search.py.

continuation patents are excluded, as they may artificially inflate measures of innovative productivity by extending the scope or duration of earlier inventions. Third, I limit the sample to patents classified under one of the following categories in the Cancer Moonshot database: *"Drugs and Chemistry"*, *"Model Systems and Animals"*, *"Cells and Enzymes"*, or *"DNA, RNA, or Protein Sequences"*. Finally, I include only those inventors for whom complete demographic and affiliation data are available—specifically, information on gender, ethnicity, and organizational affiliation—and who exhibit non-zero patenting activity over the period under study.

To define the above-mentioned tightly-knit groups, the primary task is to partition the largest component into distinct communities. For this, I employ a modularity-based algorithm, which is based on the method developed by Traag et al. [2019]. This algorithm aims to identify a partition that approximately minimizes a "modularity" criterion. In our case, the network is highly clustered, with balanced partitions consisting of several communities of roughly equal size. To refine the model estimation, I limit the analysis to communities that include at least 25 inventors. All of these communities are then pooled together to form the network under consideration.

4.4 Variables construction

IV Construction: To measure homophily among co-authors, I adopt the framework proposed by Boschma [2005], which identifies five dimensions of proximity—geographical, cognitive[2], social, institutional, and cultural—that can influence collaboration. However, given the strong correlation between geographical and institutional proximity in this study's context, I exclude the geographical dimension from the analysis.

- **Social proximity** pertains to the relationships that exist between scientists, often nurtured through prior collaborations, which reduce the uncertainty around the abilities of potential collaborators [Uzzi, 1996]. Breschi and Lissoni [2009] underscores the role of social ties, particularly past co-authorships, in facilitating future collaborations. For this study, I define social proximity by examining co-inventor relationships that occurred between $t-5$ and $t-1$, including both direct collaborations and indirect connections (e.g., shared co-inventors, with a distance of two).

- **Cultural proximity** reflects the impact of shared social norms, languages, or cultural backgrounds on collaboration. For this, I use ethnicity as a proxy for cultural proximity, leveraging the Ethnea tool.[3] This approach is consistent with previous research [Dahlander and McFarland, 2013, Freeman and Huang, 2015], which found that ethnic similarity often increases the likelihood of collaboration.

In addition to these dimensions of proximity, I consider gender as a significant factor influencing collaboration. Gender can shape co-authorship dynamics in various ways. First, the homophily hypothesis suggests that

[2] I do not consider here the impact of cognitive proximity since it leads to a huge drop in the sample size.

[3] Ethnea categorizes individuals into 26 ethnic groups, such as English, Hispanic, Chinese, German, Japanese, French, Italian, Indian, Arab, among others.

women may prefer to collaborate with other women, while men may do the same. Second, gender may introduce a potential bias, where male co-authors are favored over female ones. Finally, gender may influence collaboration decisions through risk aversion, as discussed by Ductor et al. [2023], with female inventors possibly avoiding collaborations in projects perceived as risky. The majority of inventors in the sample

Table 1: Descriptive statistics for the sample 2001-2003

	obs.	Mean	S.D	Min	Max
Weighted production	1,222	3.22	4.27	0.42	67.53
Male	1,222	0.82	0.39	0	1
Seniority (decades since first application)	1,222	0.81	0.71	0.10	3.10
Asinh (Weighted past production)	1,222	1.34	1.36	0.00	5.68
Number of patents	1,222	3.32	4.22	1	63
Science	1,222	0.76	0.42	0	1
Affiliated with a Compagny	1,222	0.86	0.35	0	1

are male (82%), which aligns with the gender composition typically observed in STEM fields. The average seniority—measured in decades since an inventor's first patent application—is 0.81, indicating that most inventors are relatively experienced. A substantial share have also published scientific articles, which is expected given the science-based nature of biomedical research. Finally, consistent with the fact that most patenting activity occurs in the private sector, the vast majority of inventors are affiliated with a company.

5 Main results

I begin my empirical analysis by showing that sharing a common cultural background is a relevant predictor of collaboration. This exercise is the formal "first step" in the approach described in Section 3.3.2. Then I present the estimates of the "second step" of the approach, described in Section 3.3.2.

5.1 Do birds of a feather flocks together?

I begin my empirical analysis by presenting the estimation results of the dyadic network formation model, which is used to instrument the adjacency matrix. This step serves as the formal first stage of the approach outlined earlier. Table (2) shows the estimation results for Model (3), incorporating all the explanatory variables identified in the literature (as discussed in the previous section).

It is instructive to begin the analysis by examining my exclusion restriction. The explanatory power of the exclusion condition—specifically, sharing a common cultural background defined here in a broad sense—is confirmed by a significant coefficient with the expected sign. This suggests that two inventors with similar cultural background tend to have a higher collaboration intensity on average. This finding aligns with existing literature that highlights the importance of shared cultural background in driving collaboration formation.

Turning to the social dimension, we observe that being closer in the network (at distance 1 or 2) facilitates collaboration formation, with past co-inventors sharing proximity significantly increasing collaboration intensity. This result aligns with prior studies linking social proximity to higher collaboration intensity. A plausible explanation is that proximity within the network provides valuable information about the quality of potential co-inventors, especially regarding their productivity.

Regarding the institutional dimension, the expected homophily effect holds true for both components considered. We find that sharing a similar institutional framework positively impacts collaboration intensity. One explanation for this is that common institutional environments reduce the costs associated with adjusting to differing operational norms, making collaboration smoother. Additionally, institutional affiliation appears to enhance collaboration intensity, as internal collaborations carry lower risks of sensitive knowledge leakage to competitors. In other words, collaborations within the same organization help preserve confidentiality.

6 On the origin of peer-effects among inventors

Regarding the influence of inventor characteristics on research output, I find that the cumulative novelty produced by an inventor'sin the past is a positive and significant predictor of productivity. This result is consistent with prior studies [Singh and Fleming, 2010, Ductor, 2015], which suggest that past impact, as measured by past production weighted by their novelty, correlates strongly with sustained inventive activity. In contrast, seniority, measured by the number of decades since an inventor's first patent application, has a negative effect on research output. This finding echoes Ductor [2015], who shows that longer career duration is associated with reduced productivity, and aligns with the life-cycle effects documented by Levin and Stephan [1991], which highlight a general decline in scientific output over the course of a career.

To understand the mechanisms driving peer effects among inventors, I examine the statistical significance of the parameters ϕ_1 and ϕ_2 in the general network model.

The results suggest that peer effects among inventors arise effectively from two distinct channels: strategic complementarity and conformity.

These findings highlight important methodological caveats in the empirical analysis of peer effects. Because peer influence is multifaceted, overlooking one significant mechanism can lead to biased estimation of the other. For instance, attributing all peer influence to strategic complementarity, while ignoring conformity, may misrepresent the underlying drivers of inventor behavior and inflate the estimated effect.

To further assess the robustness of the model and the nature of peer effects, I compare the model's predictions with those derived from standard network centrality measures that are not explicitly supported by

Table 2: Logistic regression of link formation.

Same gender	0.077^{*}
	(0.040)
Asinh(Difference in Patenting experience)	-0.066^{***}
	(0.019)
Asinh(Difference of hit Patents)	0.016
	(0.012)
Asinh(Difference of hit science)	-0.032^{***}
	(0.007)
Same institution	1.393^{***}
	(0.054)
Same Ethnicity	0.226^{***}
	(0.037)
Past Co-inventor	2.129^{***}
	(0.111)
Past Common Co-inventor	0.300^{***}
	(0.084)
Observations	30,205
Log Likelihood	−11,068.490
Akaike Inf. Crit.	22,200.980
McFadden Pseudo R^2	0.47
community FE	Yes

Note Results for Model (3) of the article are displayed. The dependent variable is defined as the existence of collaboration betwen inventor i and inventor j. The independent variables capture differences in characteristics between i and j. A precise definition of the variables at the individual level can be found in the previous section. Standard errors are reported in parentheses. An intercept is included. $*$, $**$, and $***$ indicate statistical significance at the 10%, 5%, and 1% levels.

the model. This comparison provides insight into how the analysis might change if these alternative metrics were included in the estimation.

Among the most widely used centrality measures, closeness and betweenness centrality offer different perspectives on a scientist's position in the network. Closeness centrality captures how quickly a researcher can access or disseminate information within the network, emphasizing proximity to all other nodes. In contrast, betweenness centrality, as defined by Freeman et al. [2002], measures the extent to which an inventor acts as a bridge between otherwise unconnected nodes, reflecting their potential to mediate the flow of information and ideas. Inventors with high betweenness centrality often serve as connectors across disparate groups, granting them access to diverse knowledge streams that can enhance their productivity and influence.

Table 3: Results for the period 2001-2003

	Dependent variable:		
	Number of patents weighted by novelty		
	2SLS-1	2SLS-2	2SLS-2
ϕ_1	0.03***	0.021***	0.038**
	(0.0017)	(0.0034)	(0.0179)
ϕ_2	0.054	0.813***	0.65***
	(0.0732)	(0.1813)	(0.2507)
Seniority	-0.192***	-0.161***	-0.212***
	(0.0354)	(0.0418)	(0.0697)
Asinh(Weighted past production)	0.255***	0.252***	0.288***
	(0.0205)	(0.0237)	(0.0466)
Male	0.084**	0.101**	0.11**
	(0.0399)	(0.046)	(0.0556)
Closeness Centrality			-0.649
			(59.7931)
Betweeness centrality			-0.019
			(0.0177)
Eigenvector centrality			-0.861
			(0.9981)
Cragg-Donald Wald F statistic	58.4984	35.9673	29.4
OIR test p-value	0.1304	0.18	0.2084
Community fixed effect	Yes	Yes	Yes

Notes: Estimates are derived from three models: 2SLS-1, which treats the adjacency matrix $\mathbf{G}$ as exogenous; 2SLS-2, which uses the predicted adjacency matrix $\hat{\mathbf{G}}$ as an instrument for the observed adjacency matrix $\mathbf{G}$. For columns 1-3, the dependent variable is the total number of patents over the study period, weighted by their novelty. The following dummy variable is included: Male. Inventor's experience is captured by the variable Decades after the first patent application. The past production of an inventor is captured by the number of past patents weighted by their novelty transformed with the inverse hyperbolic sine. Heteroscedasticity-robust standard errors are reported in parentheses. Statistical significance is indicated by $*$, $**$, and $***$ for the 10%, 5%, and 1% levels, respectively.

I also consider eigenvector centrality, which can be interpreted as a special case of Katz-Bonacich centrality where the decay factor equals the dominant eigenvalue of the adjacency matrix. This metric captures the extent to which a scientist is connected to other well-connected individuals without taking into account heterogeneity due to observables characteristics.

If these centrality measures significantly predict research output, it may indicate that information contagion, rather than strategic complementarity or conformity, plays a dominant role in shaping inventor productivity. Distinguishing between these mechanisms is critical for understanding the sources of peer influence in academic and innovative environments, and for designing policies that effectively foster collaborative innovation. The estimates in column 3 of Table (3) support the proposed microfoundation while rejecting the information contagion hypothesis.

7 Discussion and Conclusion

In this paper, I presented a theory of peer effects to study inventive productivity when inventors care about the behaviour of their collaborator. The theory predicts that inventor's output increase in their Katz-Bonacich centralities, a standard measure of centrality in networks.

I then tested the predictions of our model using a unique dataset of inventors in the field of cancer research. I explored the role of network position for peer effects in inventive productivity in this. I then demonstrated that, after controlling for observable characteristics and unobservable network specific factors, an inventor's position in a network (as measured by her Katz – Bonacich centrality) is a key determinant of their productivity level.

My results validate the importance of network in shaping inventors productivity, producing new evidence that the network position plays an important role in the productivity of inventors. From a general perspective, these results are indicative, in both significance and magnitude, of the relevance of network externalities in the innovation productivity. Uncovering the influence of immediate collaborators and, more generally, the network structure on productivity has important implications, not just for the production of innovation, but also for long-run growth potential. The role of peer learning has been featured in the endogenous growth literature [Romer, 1990]. My results gives an empirical argument toward the roots of peer effects, by unpacking their microfoundations by providing evidence of strategic complementarity and conformity.

There is number of possible extensions of this work. First, It would be interesting to unpack the heterogeneity of the peers effects not only based on the network but also making them a function of idiosyncratic characteristics. Second, in this work, I consider a utility function where peer effects are the results of strategic complementarity and conformity defined as the incentives for an inventor to match the average productivity of her collaborator. It would be interesting to consider alternative definition of conformity such as the willingness to conform to the most productive collaborator. Finally, since some inventors are also involved in scientific production, one promising avenue left for further research is the study of the interraction between efforts spent in innovation and scientific production in parralel.

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A APPENDIX

Table 4: Combined regression results 2004-2006

Table 5: **Logistic regression of link formation 2004-2006**

Same gender	0.012
	(0.041)
Asinh(Difference in Patenting experience)	-0.041^{**}
	(0.021)
Asinh(Difference of hit Patents)	0.015
	(0.013)
Asinh(Difference of hit science)	-0.040^{***}
	(0.008)
Same institution	1.627^{***}
	(0.077)
Same Ethnicity	0.119^{***}
	(0.039)
Past Co-inventor	1.669^{***}
	(0.078)
Past Common Co-inventor	0.354^{***}
	(0.059)
Observations	21,628
Log Likelihood	−8,903.386
Akaike Inf. Crit.	17,862.770
McFadden Pseudo R^2	0.40
community FE	Yes

Note Results for Model (3) of the article are displayed. The dependent variable is defined as the existence of collaboration betwen inventor i and inventor j. The independent variables capture differences in characteristics between i and j. A precise definition of the variables at the individual level can be found in the previous section. Standard errors are reported in parentheses. An intercept is included. $*$, $**$, and $***$ indicate statistical significance at the 10%, 5%, and 1% levels.

Table 6: **Results for the period 2004-2006**

	Dependent variable:		
	Number of patents weighted by novelty		
	2SLS-1	2SLS-2	2SLS-2
ϕ_1	0.026***	0.027***	0.035**
	(0.0021)	(0.0071)	(0.0168)
ϕ_2	-0.28**	0.715***	0.851***
	(0.114)	(0.2049)	(0.1918)
Seniority	-0.163***	-0.176***	-0.255***
	(0.044)	(0.0447)	(0.0922)
Asinh(Weighted past production)	0.232***	0.252***	0.287***
	(0.025)	(0.0308)	(0.0482)
Male	0.006	-0.049	-0.043
	(0.0479)	(0.0487)	(0.0581)
Closeness Centrality			-50.014
			(49.0123)
Betweeness centrality			0.001
			(0.0117)
Eigenvector centrality			-4.046
			(4.3869)
Cragg-Donald Wald F statistic	31.7988	17.1366	14.9761
OIR test p-value	0.0046	0.1098	0.3667
Community fixed effect	Yes	Yes	Yes

Notes: Estimates are derived from three models: 2SLS-1, which treats the adjacency matrix $\mathbf{G}$ as exogenous; 2SLS-2, which uses the predicted adjacency matrix $\hat{\mathbf{G}}$ as an instrument for the observed adjacency matrix $\mathbf{G}$. For columns 1-3, the dependent variable is the total number of patents over the study period, weighted by their novelty. The following dummy variable is included: Male. Inventor's experience is captured by the variable Decades after the first patent application. The past production of an inventor is captured by the number of past patents weighted by their novelty transformed with the inverse hyperbolic sine. Heteroscedasticity-robust standard errors are reported in parentheses. Statistical significance is indicated by $*$, $**$, and $***$ for the 10%, 5%, and 1% levels, respectively.

Table 7: Combined regression results 1997-1999

Table 8: **Logistic regression of link formation 1997-1999**

Same gender	0.037
	(0.055)
Asinh(Diff. in Patenting exp.)	−0.096***
	(0.030)
Asinh(Diff. of hit Patents)	0.100***
	(0.017)
Asinh(Diff. of hit science)	−0.028***
	(0.010)
Same institution	1.678***
	(0.078)
Same Ethnicity	0.185***
	(0.053)
Past Co-inventor	1.752***
	(0.125)
Past Common Co-inventor	0.562***
	(0.099)
Observations	13,045
Log Likelihood	−5,010.581
AIC	10,069.160
McFadden R^2	0.45
Community FE	Yes

Note Results for Model (3) of the article are displayed. The dependent variable is defined as the existence of collaboration betwen inventor i and inventor j. The independent variables capture differences in characteristics between i and j. A precise definition of the variables at the individual level can be found in the previous section. Standard errors are reported in parentheses. An intercept is included. $*$, $**$, and $***$ indicate statistical significance at the 10%, 5%, and 1% levels.

Table 9: **Results for the period 1997-1999**

	Dependent variable:		
	Number of patents weighted by novelty		
	2SLS-1	2SLS-2	2SLS-2
ϕ_1	0.039***	0.046***	0.046***
	(0.0024)	(0.0104)	(0.0176)
ϕ_2	0.04	0.714**	0.584**
	(0.0696)	(0.2852)	(0.2668)
Seniority	−0.241***	−0.18**	−0.138*
	(0.0672)	(0.0745)	(0.0751)
Asinh(Weighted past production)	0.337***	0.298***	0.262***
	(0.034)	(0.0452)	(0.0523)
Male	0.118*	0.177**	0.161**
	(0.0686)	(0.0871)	(0.0776)
Closeness Centrality			35.291
			(65.3997)
Betweenness Centrality			0.007
			(0.0113)
Eigenvector Centrality			−1.584
			(1.3137)
Cragg-Donald F-stat	40.47	15.10	12.42
OIR test p-value	0.0101	0.4807	0.3775
Community FE	Yes	Yes	Yes

Notes: Estimates are derived from three models: 2SLS-1, which treats the adjacency matrix $\mathbf{G}$ as exogenous; 2SLS-2, which uses the predicted adjacency matrix $\hat{\mathbf{G}}$ as an instrument for the observed adjacency matrix $\mathbf{G}$. For columns 1-3, the dependent variable is the total number of patents over the study period, weighted by their novelty. The following dummy variable is included: Male. Inventor's experience is captured by the variable Decades after the first patent application. The past production of an inventor is captured by the number of past patents weighted by their novelty transformed with the inverse hyperbolic sine. Heteroscedasticity-robust standard errors are reported in parentheses. Statistical significance is indicated by $*$, $**$, and $***$ for the 10%, 5%, and 1% levels, respectively.